LINE FOLLOWER

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Members:5

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INTRODUCTION

A line following robot is an autonomous mobile system designed to detect and follow a predefined path, typically represented by a black line on a contrasting surface. Such robots play a significant role in the field of automation and robotics due to their simplicity, efficiency, and wide range of applications. The robot operates by continuously sensing the position of the line and adjusting its movement to remain aligned with the track.

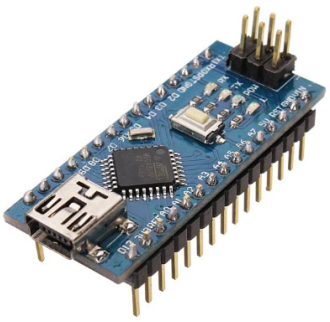
A five-sensor IR array is employed to detect the line accurately, enabling real-time correction and smooth motion. The robot's movement is driven by DC gear motors, which are controlled using a TB6612FNG motor driver that provides efficient speed and direction control through PWM signals. A buck converter is utilized to regulate the battery voltage and supply a stable 5V power to the controller and peripherals.

Additionally, a TCS34725 color sensor is integrated using the I²C interface to enhance sensing capability, and an LCD module is used to display the distance traveled by the robot in real time. Proper tuning of the control parameters ensures improved tracking accuracy and performance. The developed system demonstrates reliable autonomous operation and can be applied in industrial material handling systems, household automation, and guided navigation applications such as museum tour assistance.

COMPONENTS

1.Arduino nano board R3 with CH340

The Arduino Nano R3, built around the ATmega328P, serves as the compact central control unit for this application. It is responsible for executing the line-following algorithm and processing inputs from sensors. The inclusion of the CH340 USB interface simplifies programming and communication.



2.600 RPM 6V N20 DC Micro Metal Gear Motor

This DC gear motor provides high speed and sufficient torque required for smooth robot movement. Operating at 6V with a speed of 600 RPM, it ensures fast response and precise control. The metal gearbox enhances durability and efficiency. It is commonly used in small mobile robots.

3.TB6612FNG Dual Motor Driver Module

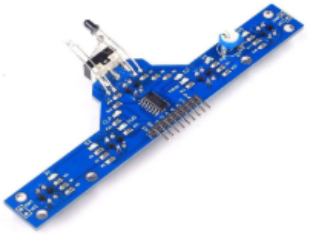
The TB6612FNG motor driver is used to control the speed and direction of two DC motors independently. It supports PWM-based control and delivers up to 1A continuous current per channel.

4.Mini MP1584 DC–DC Buck Converter Module

The MP1584 buck converter is used to step down the battery voltage to a stable 5V output.The adjustable output makes it flexible for different voltage requirements. It improves power efficiency and system stability.

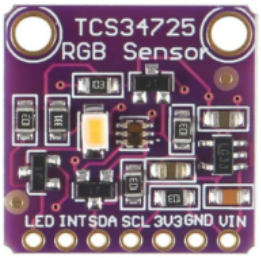
5.IR Sensor Array Line Tracking Module

The IR sensor array is used to detect the black line on a contrasting surface. Multiple sensors provide accurate position feedback, allowing precise path correction. The module is compatible with Arduino and supports fast response.



6.TCS34725 Color Sensor

The TCS34725 is a high-precision RGB color sensor interfaced using the I²C protocol. It enables accurate color detection under varying lighting conditions. In this project, it enhances sensing capability for advanced line or surface identification. An onboard LED improves measurement consistency.

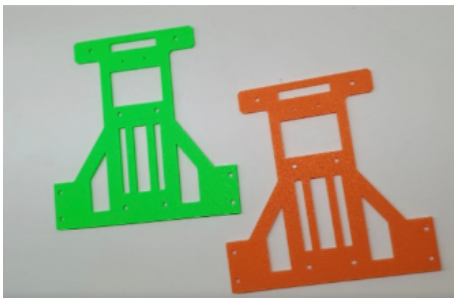


7.Rechargeable Battery

The rechargeable battery serves as the primary power source for the robot. It provides sufficient energy for motors, sensors, and control circuitry.

8.Chassis

The chassis forms the mechanical framework of the robot. It holds all components firmly in place and ensures proper weight distribution. A well-designed chassis improves stability and maneuverability. It also protects components from mechanical damage



9.Wheels

Wheels are attached to the DC motors to convert rotational motion into linear movement. They provide traction and smooth navigation along the track. Proper wheel alignment ensures accurate turning and stability.

10. ON OFF PUSH BUTTON SWITCH SELF LOCKING

The self-locking ON/OFF push button switch is used to control the power supply of the robot. Once pressed, it remains in the ON state until pressed again to turn OFF, eliminating the need for continuous holding

APPROXIMATED COSTS FOR THE EQUIPMENTS

SR NO.	EQUIPMENT	QUANTITY	AMOUNT
1.	Arduino nano board R3 with CH340	1	Rs.199
2.	600 RPM 6v N20 DC micro metal gear box motor	2	Rs.275/per
3.	TB6612FNG Dual Motor Driver Module	1	Rs.169
4.	Mini MP1584 DC–DC Buck Converter Module	1	Rs.69
5.	IR Sensor Array Line Tracking Module	1	Rs.249
6.	TCS34725 Color Sensor	1	Rs.249
7.	Rechargeable Battery	2	Rs.79
8.	Chassis	1	Rs.169
9.	Wheels	2	Rs.39
10.	Switch	1	Rs.10

CONNECTIONS AND WORKING PRINCIPLE

The line follower robot employs a TB6612FNG dual motor driver module to control two DC gear motors, enabling independent control of the left and right wheels and facilitating differential steering. The motor terminals of the TB6612FNG driver are directly connected to the respective left and right DC motors through output channels A01–A02 and B01–B02.

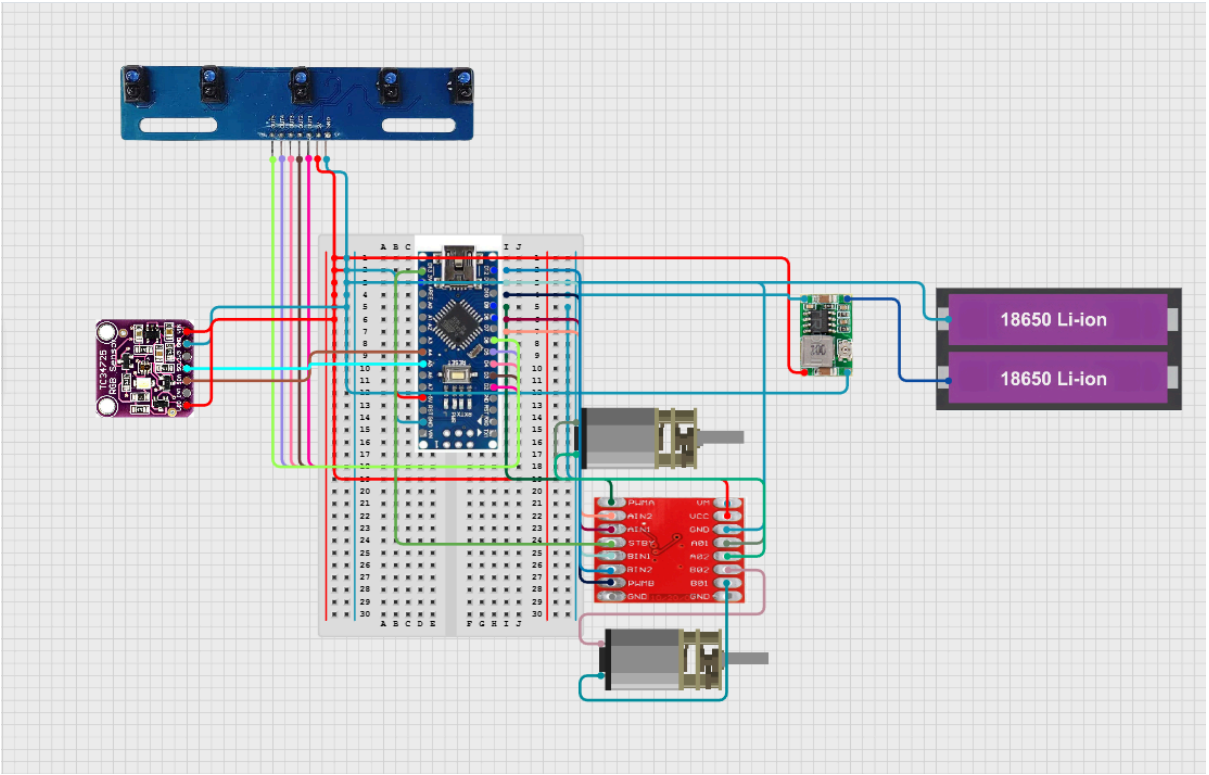
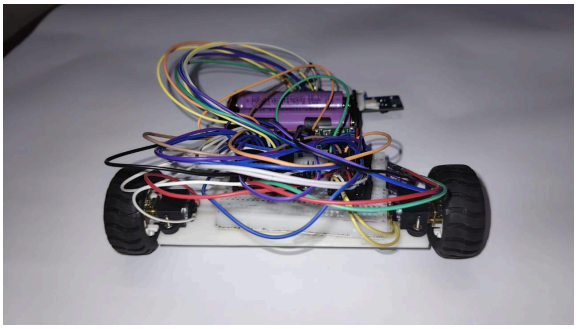
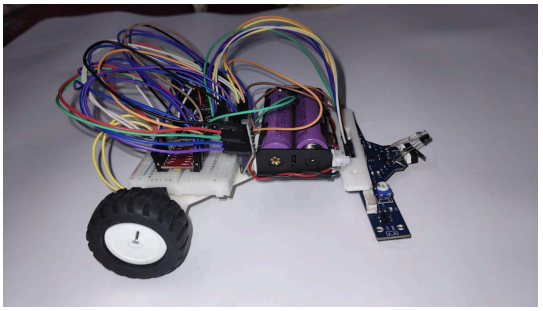
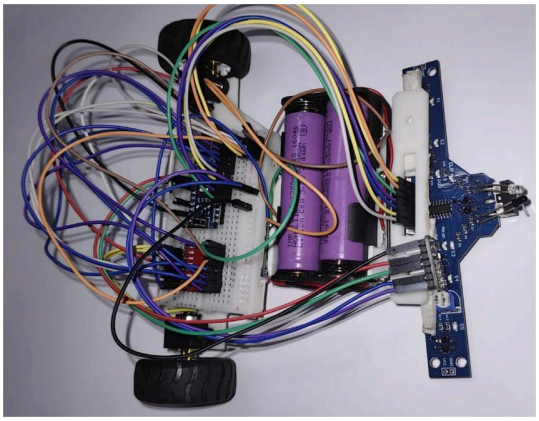
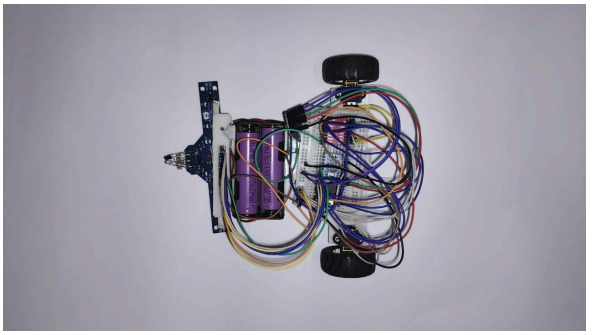
A rechargeable battery is used as the primary power source for the system. The battery voltage is supplied to a buck converter, which steps down the voltage to a regulated 5V output. This regulated supply powers the Arduino Nano, the IR sensor array, the color sensor, and the logic supply (VCC) of the motor driver. The motor supply pin (VM) of the TB6612FNG is connected directly to the battery to provide sufficient power to the motors. A common ground is maintained across all components to ensure stable and reliable operation.

The control pins of the motor driver are interfaced with the Arduino Nano. Pulse Width Modulation (PWM) pins PWMA and PWMB are connected to digital pins D9 and D10 respectively to control motor speed, while direction control pins AIN1, AIN2, BIN1, and BIN2 are connected to digital pins D8, D7, D11, and D12. The standby (STBY) pin of the motor driver is connected to Arduino pin D13 to enable or disable motor operation.

Line detection is achieved using a five-channel IR sensor array, where each sensor output is connected to individual digital input pins (D2–D6) of the Arduino Nano. The sensor module is powered using the regulated 5V supply from the buck converter, while the CLP and CLEAR pins are left unconnected as they are not required for this application. Additionally, a TCS34725 color sensor is interfaced with the Arduino via the I²C protocol using pins A4 (SDA) and A5 (SCL), with its onboard LED connected to the 5V supply.

During operation, the IR sensors continuously monitor the surface beneath the robot and detect the position of the black line relative to the robot. The Arduino Nano processes these sensor signals and determines the required direction and speed adjustments. Based on this decision, appropriate PWM signals are generated and applied to the motor driver, which regulates the speed and direction of the DC motors. By varying the relative speeds of the left and right motors, the robot is able to follow the line smoothly and accurately. The coordinated operation of the sensors, controller, and motor driver enables reliable autonomous navigation along the predefined path.

CIRCUIT DIAGRAM



ROBOT IMAGES